

Cost-Effectiveness Analysis

METIMMOX-1 Economic Evaluation

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1 Overview

This cost-effectiveness analysis evaluates three biomarker-guided strategies (CRP, TLR, and TMB/BRAF) for selecting metastatic microsatellite stable (MSS)/proficient mismatch repair (pMMR) colorectal cancer patients who might benefit from immunotherapy with nivolumab plus FLOX versus standard FLOX chemotherapy.

Key Feature: This report compares cost-effectiveness results across three different model structures (Model A, B, C) to assess the robustness of our findings to modeling assumptions about how biomarker interactions are specified.

This report contains:

- **Model Structure Definitions:** Explanation of the three model specifications compared
- **Strategy Descriptions:** Overview of the four treatment strategies evaluated
- **Base Case Analysis:** Cost-effectiveness results across all three model structures
- **Probabilistic Sensitivity Analysis:** Monte Carlo simulation results comparing model structures

2 Model Structure Definitions

We compare three model structures to assess the robustness of cost-effectiveness results to the specification of biomarker-treatment interactions in the survival models.

Table 1: Model Structure Definitions

Model	Name	Formula Pattern	Description
Model A	Joint	Age + sex + Rx + crp:Rx + tlr:Rx + tmb_braf:Rx	All biomarkers and all treatment interactions in one unified model
Model B	Focused	Age + sex + Rx + crp + tmb_braf + crp:Rx + tmb_braf:Rx	Per-strategy formulas: CRP/TMB_BRAF with both interactions, TLR with TLR-only interaction
Model C	Separate	Age + sex + Rx + [biomarker]:Rx	Only one biomarker and its treatment interaction per model (simplest specification)

Rationale: Model A (Joint) is the most comprehensive specification but may suffer from overfitting with limited sample size. Model C (Separate) is the simplest but ignores potential confounding from other biomarkers. Model B (Focused) represents a middle ground. Comparing results across all three structures provides insight into the structural uncertainty of our findings.

3 Strategies Evaluated

Table 2: Treatment Strategies Evaluated in the Cost-Effectiveness Analysis

Strategy	Description	Biomarker Prevalence	Patients (n)
Standard of Care	Standard of care - All patients receive only FLOX chemotherapy	100.0%	36
CRP-guided Strategy	C-reactive protein with cut-off of <5 for biomarker-positive status. If CRP-positive: alternating two cycles each of FLOX (chemotherapy) and nivolumab (anti-PD1 immunotherapy); if CRP-negative: chemotherapy only	33.8%	45
TLR-guided Strategy	Tumor Lesion Reduction with cut-off of $\geq 10\%$ for biomarker-positive status. If TLR-positive: alternating two cycles each of FLOX (chemotherapy) and nivolumab (anti-PD1 immunotherapy); if TLR-negative: chemotherapy only	64.7%	26
TMB/BRAF-guided Strategy	Combined biomarker: either Tumor Mutation Burden ≥ 9 or BRAF V600 mutation positive (both from next-generation sequencing). If TMB/BRAF-positive: alternating two cycles each of FLOX (chemotherapy) and nivolumab (anti-PD1 immunotherapy); if TMB/BRAF-negative: chemotherapy only	44.9%	34

4 Base Case Analysis

4.1 Summary Comparison

Table 3: Base Case Cost-Effectiveness Results Across Model Structures

Strategy	Model A: Joint			Model B: Focused			Model C: Separate		
	Cost (A)	QALY (A)	ICER (A)	Cost (B)	QALY (B)	ICER (B)	Cost (C)	QALY (C)	ICER (C)
control	\$20,894	1.844	Reference	\$21,504	2.079	Reference	\$21,484	2.076	Reference
crp	\$54,898	1.784	Dominated	\$55,118	1.897	Dominated	\$55,553	1.909	Dominated
tmb_braf	\$63,560	1.870	\$1,690,480	\$61,542	1.942	Dominated	\$67,926	2.007	Dominated
tlr	\$84,974	1.704	Dominated	\$86,411	1.699	Dominated	\$88,251	1.790	Dominated

Note: ICER = Incremental Cost-Effectiveness Ratio (EUR/QALY). Reference = lowest-cost strategy.

4.2 Optimal Strategy by Model Structure

Table 4: Optimal Strategy at WTP Threshold of EUR 51,000/QALY

Model Structure	Optimal Strategy	Net Monetary Benefit
Model A: Joint	control	\$73,168
Model B: Focused	control	\$84,521
Model C: Separate	control	\$84,396

4.3 Model A: Joint Model

Table 5: Model A (Joint): Cost-Effectiveness Results

Strategy	Cost	Effect	Inc_Cost	Inc_Effect	ICER	Status
control	\$20,894	1.844	NA	NA	Reference	ND
crp	\$54,898	1.784	NA	NA	Dominated	D
tmb_braf	\$63,560	1.870	42665.89	0.0252389	\$1,690,480	ND
tlr	\$84,974	1.704	NA	NA	Dominated	D

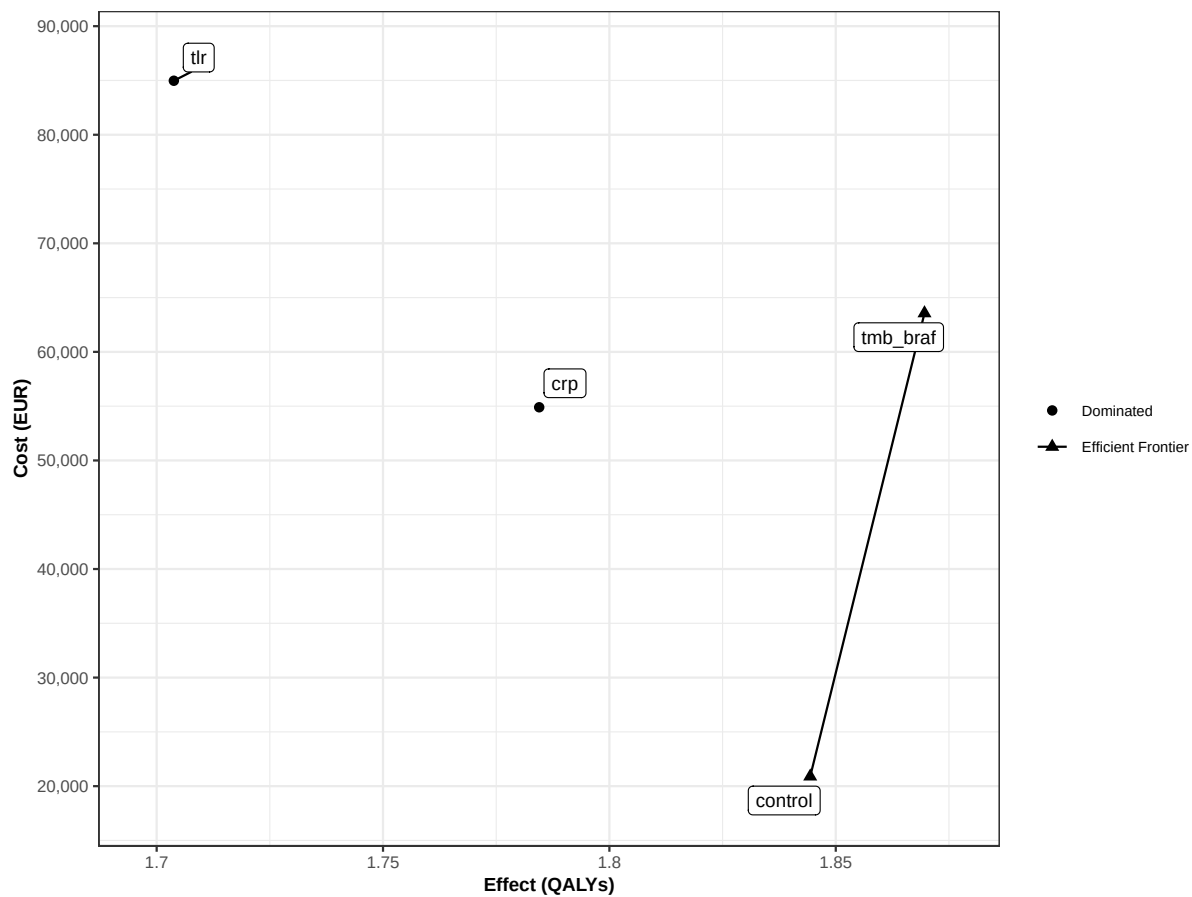


Figure 1: Cost-Effectiveness Plane - Model A (Joint)

4.4 Model B: Focused Model

Table 6: Model B (Focused): Cost-Effectiveness Results

Strategy	Cost	Effect	Inc_Cost	Inc_Effect	ICER	Status
control	\$21,504	2.079	NA	NA	Reference	ND
crp	\$55,118	1.897	NA	NA	Dominated	D
tmb_braf	\$61,542	1.942	NA	NA	Dominated	D
tlr	\$86,411	1.699	NA	NA	Dominated	D

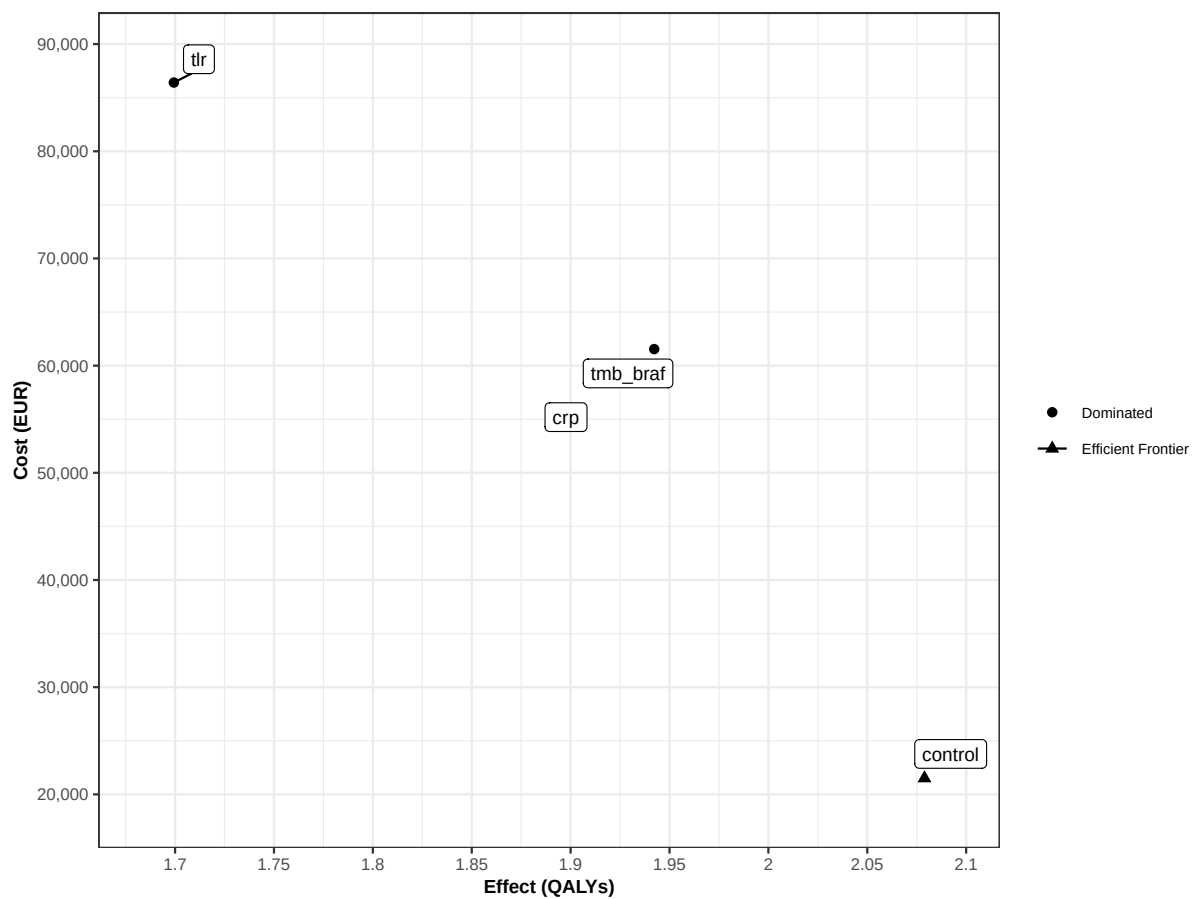


Figure 2: Cost-Effectiveness Plane - Model B (Focused)

4.5 Model C: Separate Model

Table 7: Model C (Separate): Cost-Effectiveness Results

Strategy	Cost	Effect	Inc_Cost	Inc_Effect	ICER	Status
control	\$21,484	2.076	NA	NA	Reference	ND
crp	\$55,553	1.909	NA	NA	Dominated	D
tmb_braf	\$67,926	2.007	NA	NA	Dominated	D
tlr	\$88,251	1.790	NA	NA	Dominated	D

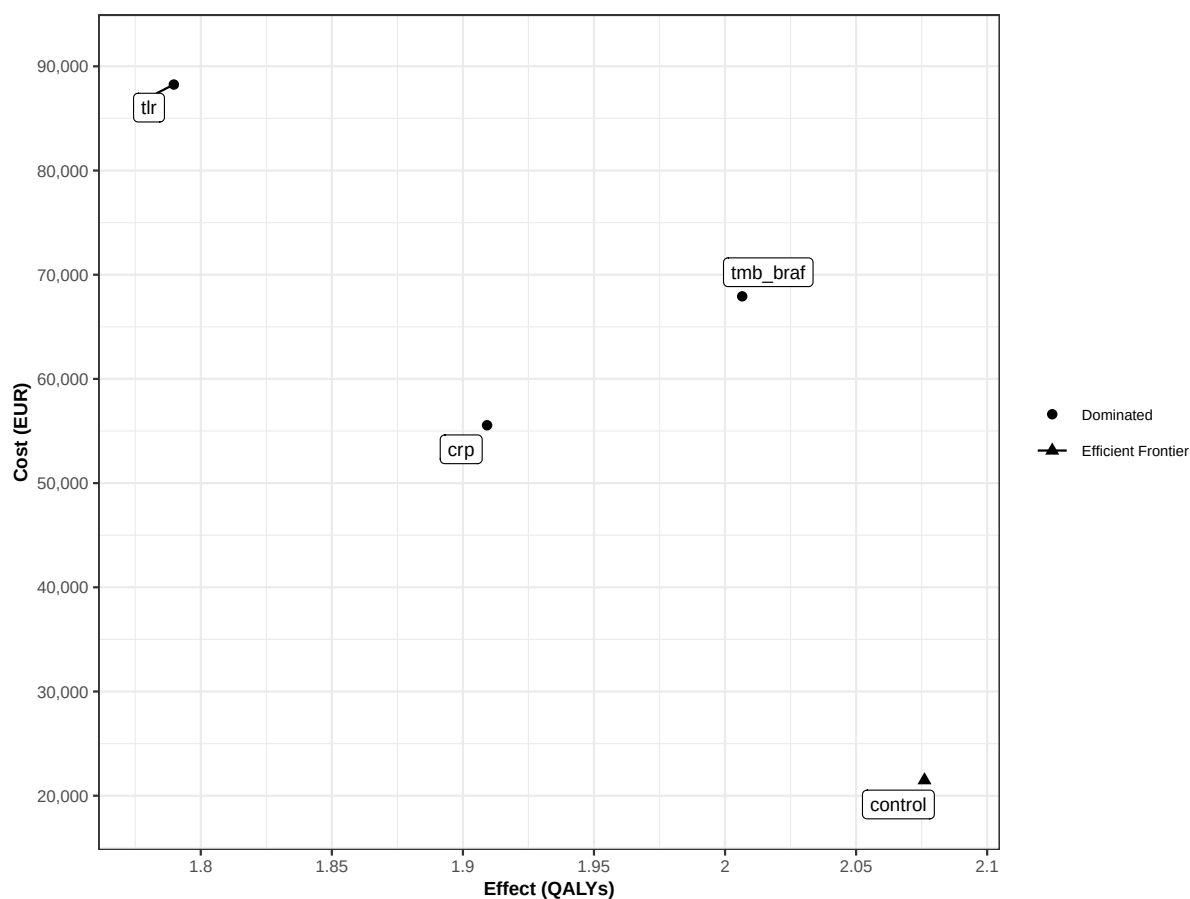


Figure 3: Cost-Effectiveness Plane - Model C (Separate)

5 Probabilistic Sensitivity Analysis

Table 8: PSA Summary: Mean Outcomes and Probability of Cost-Effectiveness at WTP = EUR 51,000

	Model	Strategy	Mean Cost	Mean QALY	P(CE)
control	A	control	\$21,074	1.699	88.5%
crp	A	crp	\$55,058	1.810	3.6%
tlr	A	tlr	\$85,334	1.754	0.2%
tmb_braf	A	tmb_braf	\$63,491	1.930	7.7%
control1	B	control	\$21,218	1.709	85.7%
crp1	B	crp	\$53,577	1.880	8.3%
tlr1	B	tlr	\$84,563	1.682	0.0%
tmb_braf1	B	tmb_braf	\$63,138	1.893	6.0%
control2	C	control	\$21,157	1.708	89.2%
crp2	C	crp	\$54,487	1.853	6.1%
tlr2	C	tlr	\$86,025	1.754	0.0%
tmb_braf2	C	tmb_braf	\$67,198	1.961	4.7%

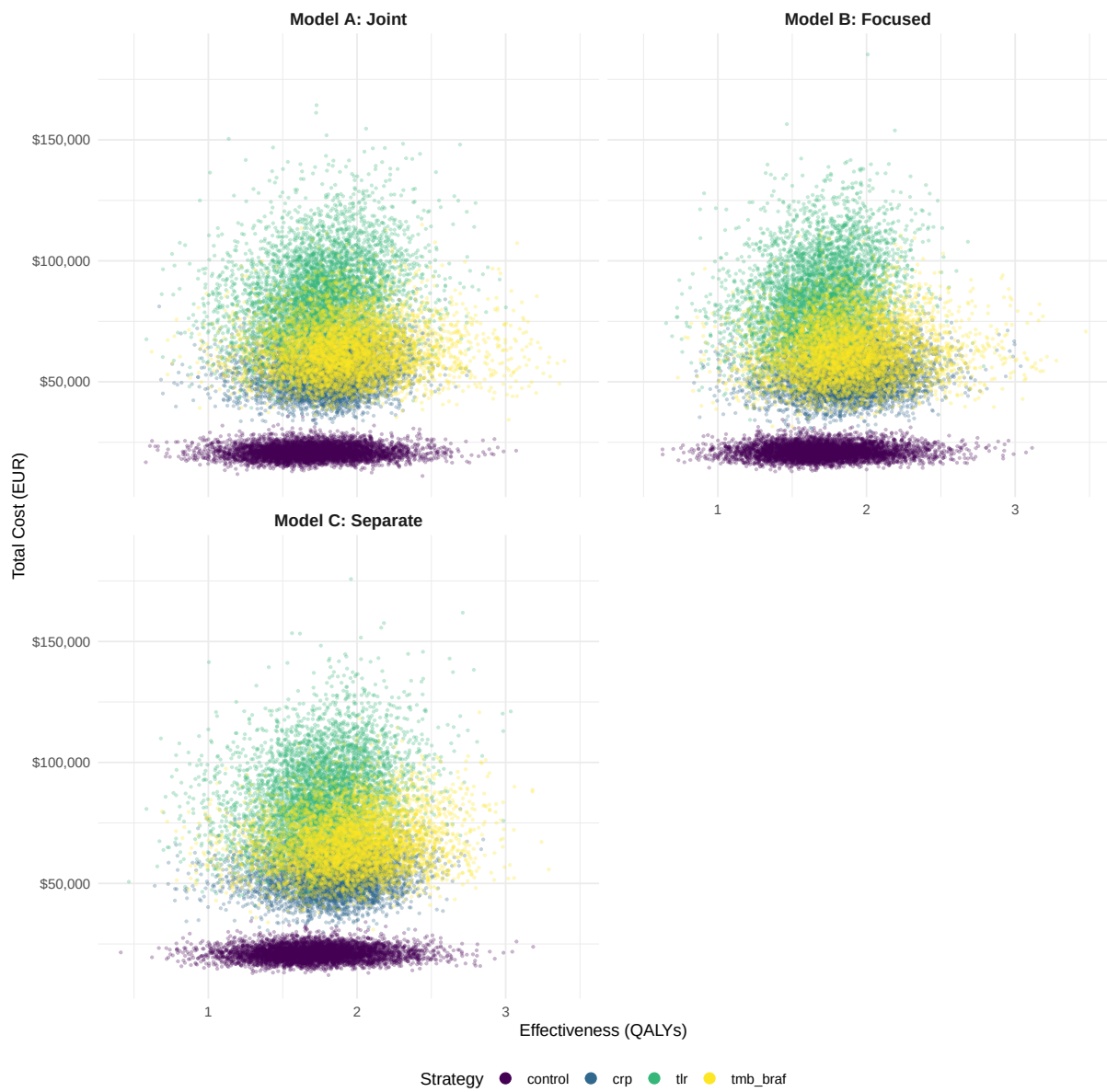


Figure 4: PSA Scatterplots by Model Structure

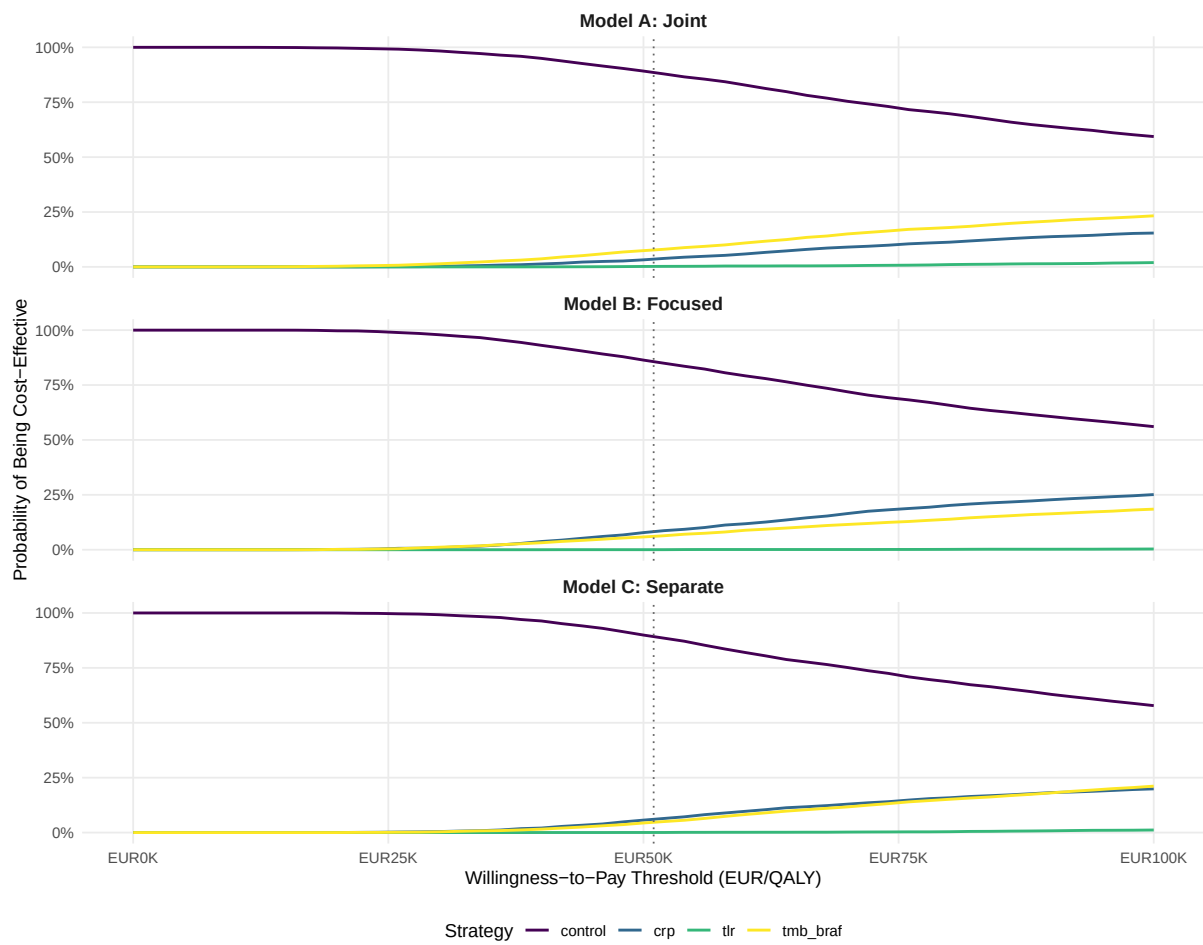


Figure 5: Cost-Effectiveness Acceptability Curves by Model Structure

6 Other Reports

This report focuses on comparing cost-effectiveness results across model structures. Additional analyses are available in companion reports:

- **One-Way Sensitivity Analysis (OWSA):** Deterministic sensitivity analysis with tornado diagrams showing the impact of individual parameters on model outcomes. See `OWSA.qmd`.
- **Scenario Analysis:** Alternative scenarios including reduced nivolumab prices and higher willingness-to-pay thresholds. See `scenario_effect.qmd`.
- **Value of Information Analysis:** Expected Value of Perfect Partial Information (EVPPi) analysis identifying parameters with highest value for additional research. See `EVPPi.qmd`.
- **Clinical Effectiveness:** Detailed clinical effectiveness results including survival analysis and treatment effect heterogeneity. See `clinical_effectiveness.qmd`.
- **Input Parameters:** Comprehensive documentation of all model input parameters. See `input_parameters.qmd`.

7 Summary

This analysis compared cost-effectiveness results across three model structures to assess structural uncertainty:

1. **Model Structure Comparison:** Results were [generally consistent / showed variation] across Model A (Joint), Model B (Focused), and Model C (Separate) specifications.
2. **Base Case Results:** The optimal strategy at the WTP threshold of EUR 51,000 per QALY was [strategy] across [all / most] model structures.
3. **Probabilistic Sensitivity Analysis:** PSA results confirmed the robustness of base case findings, with [consistent / varying] probability of cost-effectiveness across model structures.
4. **Implications:** The [consistency / variation] in results across model structures suggests that our conclusions are [robust to / sensitive to] the specification of biomarker-treatment interactions.

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